

Figure 1. Overall framework of real-hardware Nussbaum–PID control and optimization on a NIRYO NED3 Pro manipulator.



CORE IDEA: Keep the interpretable Nussbaum–PID structure, then adapt and tune the hardware-facing terms to the measured behavior of the real actuator.

Graphical Abstract. Closing the gap between theory and real hardware through hardware-aware adaptation and Optuna-guided tuning.

